CHAPTER 1

# RELATIONS AND FUNCTIONS

**CHAPTER ESSENCE AT A GLANCE:**
Properties of relations (reflexive, symmetric, transitive, equivalence classes), function types (injective, surjective, bijective), and composition.

## 4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Equivalence Relations and Partitions:** A relation on a set is an equivalence relation if and only if it satisfies reflexivity, symmetry, and transitivity simultaneously. Any equivalence relation partitions the entire underlying set into pairwise disjoint equivalence classes whose union reconstructs the complete set.

• **Injectivity and the One-to-One Criterion:** A function is injective or one-to-one when distinct domain elements always produce distinct images in the co-domain. Analytically, assuming equality of outputs forces the equality of corresponding inputs.

• **Surjectivity and Co-domain Exhaustion:** A function is surjective or onto when every element in the co-domain serves as an image for at least one domain pre-image, establishing that the functional range equals the co-domain.

• **Bijectivity and Invertibility:** A function that is both injective and surjective achieves an exact one-to-one correspondence between sets, which guarantees the existence of a unique inverse function.

## ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$R \text{ is equivalence} \Leftrightarrow \text{Reflexive, Symmetric, and Transitive}$$

*Equivalence relation requirement on arbitrary set A*

$$f(x_1) = f(x_2) \rightarrow x_1 = x_2 \quad \text{(Injectivity)}$$

*One-to-One condition: distinct elements have distinct images*

$$\text{Bijective} \Leftrightarrow \text{Injective (One-to-One)} + \text{Surjective (Range = Co-domain)}$$

*Bijective condition ensuring existence of unique inverse function*

$$[a] = \{x \in A : (x, a) \in R\}$$

*Equivalence class containing all elements related to a*

$$f(x) = \frac{ax+b}{cx+d} \rightarrow x = \frac{dy-b}{a-cy}$$

*Rational function inverse via algebraic rearrangement*

## CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** [Trap Alert: Vacuous Transitivity] If (a, b) R but there is NO element in R starting with b, the transitive condition is VACUOUSLY TRUE! Do not mark it non-transitive without a counterexample.

• **Alert:** [Trap Alert: Modulus & Greatest Integer] Modulus function f(x) = |x| and Greatest Integer function f(x) = [x] from R to R are NEITHER one-one NOR onto!

## HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

### 1. Define reflexive, symmetric, and transitive relations. Define equivalence relation. [Score: 2 Marks]

**Answer & Key Valuation Steps:**

• **[Step 1: Definitions of Relations on Set A]** 1. Reflexive: (a, a) in R for every a in A. 2. Symmetric: If (a, b) in R, then (b, a) in R for all a, b in A. 3. Transitive: If (a, b) in R and (b, c) in R, then (a, c) in R for all a, b, c in A. [1 Mark]

• **[Step 2: Equivalence Relation]** A relation R on set A is an EQUIVALENCE RELATION if and only if it is simultaneously Reflexive, Symmetric, and Transitive. Example: The relation of equality '=' or congruence of triangles. [1 Mark]

• **[Boxed Result: Equivalence Relation = Reflexive + Symmetric + Transitive]**

### 2. Define injective (one-one), surjective (onto), and bijective functions with standard examples. [Score: 2 Marks]

**Answer & Key Valuation Steps:**

• **[Step 1: Injective (One-One) Function]** A function f: X $\to$ Y is injective if distinct elements of X have distinct images under f: $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ for all $x_1, x_2$ in X. Example: f: R $\to$ R, f(x) = 2x. [1 Mark]

• **[Step 2: Surjective (Onto) & Bijective Function]** Surjective: Every element y in Y is the image of at least one element x in X (Range(f) = Co-domain Y): for every y in Y, $\exists$ x in X such that f(x) = y. Bijective: A function that is BOTH injective (one-one) and surjective (onto). A bijection is invertible ($f^{-1}$ exists). [1 Mark]

• **[Boxed Result: Bijection = One-One (Injective) + Onto (Surjective)]**

### 3. Show that the function f: R -> R defined by f(x) = 3 - 4x is a bijection. [Score: 3 Marks]

**Answer & Key Valuation Steps:**

• **[Setup]** f: R $\to$ R defined by f(x) = 3 - 4x. [0.5 Mark]

• **[Step 1: One-One (Injectivity)]** Let $x_1, x_2$ in R such that $f(x_1) = f(x_2)$. $3 - 4x_1 = 3 - 4x_2 \Rightarrow -4x_1 = -4x_2 \Rightarrow x_1 = x_2$. Hence f is one-one (injective). [1 Mark]

• **[Step 2: Onto (Surjectivity)]** Let y in R (co-domain). Set y = f(x) = 3 - 4x. 4x = 3 - y $\Rightarrow$ x = (3 - y) over 4. Since y in R, x = (3 - y) over 4 in R (domain). Check: f(x) = f((3 - y) over 4) = 3 - 4 [ (3 - y) over 4 ] = 3 - (3 - y) = y. Thus every y in co-domain has a pre-image in domain. Hence f is onto (surjective). [1 Mark]

• **[Conclusion]** Since f is both one-one and onto, f is BIJECTIVE. [0.5 Mark]

• **[Boxed Result: f(x) = 3 - 4x is a Bijective Function]**

### 4. Show that relation R in set of all lines in XY plane defined by L1 R L2 iff L1 is parallel to L2 is equivalence. [Score: 3 Marks]

**Answer & Key Valuation Steps:**

• **[Setup]** Set L of all lines in XY plane. Relation R: $L_1$ R $L_2 \Leftrightarrow L_1 \parallel L_2$. [0.5 Mark]

• **[Step 1: Three Properties]** 1. Reflexive: Every line is parallel to itself ($L_1 \parallel L_1$). Thus ($L_1$, $L_1$) in R. 2. Symmetric: If $L_1 \parallel L_2$, then $L_2 \parallel L_1$. Thus ($L_1$, $L_2$) in R $\Rightarrow$ ($L_2$, $L_1$) in R. 3. Transitive: If $L_1 \parallel L_2$ and $L_2 \parallel L_3$, then $L_1 \parallel L_3$. Thus ($L_1$, $L_2$) in R and ($L_2$, $L_3$) in R $\Rightarrow$ ($L_1$, $L_3$) in R. [2 Marks]

• **[Conclusion]** Since R is reflexive, symmetric, and transitive, R is an EQUIVALENCE RELATION. (Equivalence class of line y = 2x + 4 is the set of all parallel lines y = 2x + c, c in R). [0.5 Mark]

• **[Boxed Result: Parallelism of lines is an Equivalence Relation]**

### 5. Let f: X -> Y be a function. Define relation R on X by R = {(a, b): f(a) = f(b)}. Show that R is an equivalence relation. [Score: 3 Marks]

**Answer & Key Valuation Steps:**

• **[Setup]** f: X $\to$ Y. Relation on X: R = {(a, b) : f(a) = f(b)}. [0.5 Mark]

• **[Step 1: Reflexive]** For any a in X: f(a) = f(a) is trivially true. Hence (a, a) in R. [0.5 Mark]

• **[Step 2: Symmetric]** Let (a, b) in R $\Rightarrow$ f(a) = f(b) $\Rightarrow$ f(b) = f(a) $\Rightarrow$ (b, a) in R. [1 Mark]

• **[Step 3: Transitive]** Let (a, b) in R and (b, c) in R $\Rightarrow$ f(a) = f(b) and f(b) = f(c). Then f(a) = f(c) $\Rightarrow$ (a, c) in R. [1 Mark]

• **[Conclusion]** Since R is reflexive, symmetric, and transitive, R is an EQUIVALENCE RELATION on X. [0.5 Mark]

• **[Boxed Result: R = {(a, b) : f(a) = f(b)} is an Equivalence Relation]**

**6. Let R be a relation on Z defined by R = {(a, b): 2 divides (a - b)}. Prove that R is an equivalence relation. [Score: 4 Marks]**

**Answer & Key Valuation Steps:**

• **[Setup]** R = {(a, b) : a, b in Z and 2 divides (a - b)}. [0.5 Mark]

• **[Step 1: Reflexivity]** For any a in Z: a - a = 0 = 2 × 0, which is divisible by 2. Thus (a, a) in R for all a in Z. Hence R is reflexive. [1 Mark]

• **[Step 2: Symmetry]** Let (a, b) in R ⇒ 2 divides (a - b) ⇒ a - b = 2k for some k in Z. Then b - a = - (a - b) = - 2k = 2 (-k), where -k in Z. So 2 divides (b - a) ⇒ (b, a) in R. Hence R is symmetric. [1 Mark]

• **[Step 3: Transitivity]** Let (a, b) in R and (b, c) in R. Then $a - b = 2k_1$ and $b - c = 2k_2$ for $k_1$, $k_2$ in Z. Adding both: $(a - b) + (b - c) = 2k_1 + 2k_2 \Rightarrow a - c = 2 (k_1 + k_2)$, where $k_1 + k_2$ in Z. Thus 2 divides (a - c) ⇒ (a, c) in R. Hence R is transitive. [1 Mark]

• **[Step 4: Conclusion]** Since R is reflexive, symmetric, and transitive, R is an EQUIVALENCE RELATION. [0.5 Mark]

• **[Boxed Result: R is an Equivalence Relation on Z]**

**7. Let A = R - {3} and B = R - {1}. Consider f: A -> B defined by f(x) = (x - 2) over (x - 3). Prove that f is bijective. [Score: 4 Marks]**

**Answer & Key Valuation Steps:**

• **[Setup]** A = R - {3}, B = R - {1}. f: A → B given by f(x) = (x - 2) over (x - 3). [0.5 Mark]

• **[Step 1: Injectivity (One-One)]** Let $x_1$, $x_2$ in A such that $f(x_1) = f(x_2)$: $(x_1 - 2)$ over $(x_1 - 3) = (x_2 - 2)$ over $(x_2 - 3) \Rightarrow (x_1 - 2)(x_2 - 3) = (x_2 - 2)(x_1 - 3)$ $x_1 x_2 - 3 x_1 - 2 x_2 + 6 = x_1 x_2 - 3 x_2 - 2 x_1 + 6 - 3 x_1 - 2 x_2 = - 3 x_2 - 2 x_1 \Rightarrow - x_1 = - x_2 \Rightarrow x_1 = x_2$. Hence f is one-one. [1.5 Marks]

• **[Step 2: Surjectivity (Onto)]** Let y in B = R - {1}. Set y = (x - 2) over (x - 3): $y(x - 3) = x - 2 \Rightarrow x y - 3y = x - 2 \Rightarrow x(y - 1) = 3y - 2 \Rightarrow x =$ (3y - 2) over (y - 1). Since $y \neq 1$, x is well-defined in R. If x = 3: $3(y - 1) = 3y - 2 \Rightarrow 3y - 3 = 3y - 2 \Rightarrow -3 = -2$ (impossible). Thus x in A = R - {3}. f(x) = [ (3y-2) over (y-1) - 2 ] over [ (3y-2) over (y-1) - 3 ] = (3y - 2 - 2y + 2) over (3y - 2 - 3y + 3) = y over 1 = y. Hence f is onto. Since f is one-one and onto, f is BIJECTIVE. [2 Marks]

• **[Boxed Result: f(x) = (x - 2) over (x - 3) is a Bijective Function]**